

Validity Analysis: DISASTERVQA

Target context: Pampas Region Emergency Managers (Argentina) — Urban Flood Damage Assessment

Overall risk: **HIGH**

Deployment Context

Use case: Damage assessment provoked by natural disasters in urban areas, specifically floods. The information available consist of images and video captures from social media and sent by the affected local population.

Target population: Emergency managers from the Pampas region in Argentina. They speak Rio Platense Spanish, are born and raised in the cities they serve. No particular gender, most likely man between 30 and 45 years old.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content ■	1	Serious concern	high
Input Form ■	2	Significant gaps	high
Output Ontology ✓	3	Moderate gaps	high
Output Content ■	2	Significant gaps	high
Output Form ■	2	Significant gaps	high
Average	2.2		

■ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

DisasterVQA is a well-constructed diagnostic benchmark for evaluating VLMs on humanitarian-framework-grounded disaster VQA, but it is substantially misaligned with the Pampas urban flood damage assessment deployment context. The benchmark documentation explicitly acknowledges its status as an evaluation tool rather than a deployment-readiness measure, and disclaims operational use — a framing at direct odds with the intended use case. Concrete misalignments include: (a) input content overwhelmingly reflects South Asian and North American flood contexts, with only ~2% of the sample tagged 'South America' and none of these being floods (Critical dataset concern 1); (b) English-only prompts and outputs, incompatible with Rioplatense Spanish operational language; (c) FEMA ESF / OCHA MIRA output ontology not

calibrated to Argentine SINAGIR/AFE frameworks; (d) Qatar-based annotator pool with no Latin American regional expertise; (e) static-image-only modality omits video, a primary WhatsApp intake channel; (f) documented worst-in-benchmark performance on flood-specific tasks (open-ended 0.73, MCQ F1 0.63) precisely where the deployment needs accuracy. The benchmark does provide operationally relevant task categories (road passability, water depth grading, evacuation identification) and its social-media provenance matches the deployment input channel at a source level. However, using DisasterVQA scores as a selection criterion for Pampas flood response VLMs would risk substantial construct-irrelevant variance, content underrepresentation, and external-validity failure.

Practical Guidance

What This Benchmark Measures

DisasterVQA measures a VLM's zero-shot ability to answer English-language visual questions about ground-level disaster imagery, using humanitarian frameworks (FEMA ESF / OCHA MIRA) as the taxonomic scaffold. Its strongest dimensions for the Pampas use case are Input Ontology (score 3) and Output Ontology (score 3): the benchmark exercises task categories — road passability, water depth quantification, damage detection, evacuation identification — that overlap substantively with what Pampas emergency managers need to assess. The three-format design (binary, MCQ, open-ended) provides useful diagnostic granularity for comparing candidate models on flood-adjacent reasoning tasks.

Construct Depth

Construct depth is uneven. The benchmark probes flood reasoning deeply along the axes it does cover — water depth relative to human/vehicle references, road submersion grading, multi-object damage identification, and actionable-task recognition are all present with graduated MCQ options. However, this depth is calibrated against South Asian and North American visual and institutional contexts, English-language prompts, and Anglo/UN humanitarian vocabulary. The benchmark provides essentially no evidence about model behavior on: Argentine urban infrastructure (villas, casas chorizo, zanjones), Rioplatense Spanish prompting, SINAGIR/AFE-aligned outputs, video inputs, or Argentina-specific geographic performance. Reported flood open-ended accuracy of 0.73 and Water & Flood MCQ F1 of 0.63 indicate that even within the benchmark's own scope, flood-specific reasoning remains a weakness.

What Else You Need

For operational deployment in the Pampas region, DisasterVQA scores would need substantial supplementation: (1) an Argentine flood-imagery evaluation set drawn from historical events (2013 La Plata, 2023–2025 Bahía Blanca, 2015 RMBA) with Argentine emergency-manager ground truth (addresses IC and OC gaps); (2) Rioplatense Spanish prompt/output evaluation using bilingual QA pairs (addresses IF and OF gaps); (3) mapping of benchmark labels to SINAGIR/AFE damage classifications and PNRRD 2025–2029 priorities (addresses OO gap); (4) video-frame or short-clip evaluation using WhatsApp-typical video captures (addresses IF gap); (5) inter-annotator agreement study between original annotators and Argentine emergency managers on a stratified sample (addresses OC gap); (6) informal-settlement (villa/asentamiento) recognition subset (addresses IO and IC gaps).

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: DISASTERVQA | **Context:** Pampas Region Emergency Managers (Argentina) — Urban Flood Damage Assessment

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The task taxonomy covers many categories directly relevant to Pampas urban flood assessment — built environment damage, access & inaccessibility, hazard type & severity, water & flood characteristics, and rescue operations [\[Q9, Q50, Q51\]](#) — and the dataset includes operationally central question types (road drivability, water depth grading, sandbag/mitigation detection) confirmed empirically [\[D3, D4, D7, D8\]](#). However, the taxonomy is explicitly grounded in FEMA ESF and OCHA MIRA frameworks [\[Q9, Q38\]](#), not in Argentine SINAGIR/AFE frameworks that govern Pampas emergency management [\[WEB-4, WEB-5, WEB-6\]](#). Categories relevant to Argentine operations — informal-settlement (villa/asentamiento) recognition, zanjón/underpass classification, evacuation-route viability toward specific refugios — are not represented as MCQ options in the sampled data [\[D22, D23, D30\]](#). The benchmark is also explicitly framed as evaluation-only, not deployment [\[Q112, Q114\]](#), which the use case violates. Ontology is broadly plausible but not calibrated to Argentine operational logic.

ID	Evidence
Q9	"The questions are grounded in established humanitarian response frameworks (e.g., FEMA ESF (FEMA 2025), OCHA MIRA (UN OCHA 2025), etc.) and cover tasks such as identifying built environment damage, determining hazard type and severity, evaluating accessibility, and assessing movement restrictions and controls, making the dataset highly relevant for real-world disaster response applications." (p.1)
Q38	"We employed GPT-4o to categorize each verified Q&A pair based on definitions from several frameworks, including the FEMA Emergency Support Functions (ESF) (FEMA 2025) and OCHA MIRA (UN OCHA 2025)." (p.3)
Q50	"From an operational perspective, the dataset emphasizes information needs critical to humanitarian response (ref. Figure 3.c): Built Environment Damage (722 images, 1,018 questions) and Hazard Type & Severity (710 images, 912 questions) are the most prevalent categories, followed by Population/Asset Presence & Exposure (604 images, 870 questions)." (p.4)
Q51	"Access & Inaccessibility, central to routing and logistics, accounts for 410 images paired with 511 questions, while less frequent categories such as Utilities Status & Damage, Search and Rescue, and Debris Clearance & Earthmoving capture high-impact, mission-critical decision-making scenarios." (p.4)
Q112	"DisasterVQA is designed as an evaluation benchmark for assessing vision–language perception and reasoning in disaster scenarios, rather than as a dataset for model training or deployment." (p.8)
Q114	"DisasterVQA is therefore best understood as a controlled diagnostic tool for identifying strengths, failure modes, and research gaps in current vision–language models, rather than as a proxy for real-world disaster response performance." (p.8)
D3	"Is this road drivable? Answer with Yes or No." disaster_type=flood, region=null — Core accessibility question directly relevant to deployment; no geographic context attached
D4	"What is the approximate water level shown on the gauge? Select all that apply." region=South Asia, disaster_type=flood — Fine-grained water level quantification; operationally relevant for Pampas flood depth assessment
D7	"What is the level of floodwater compared to the height of the people visible in the image?" region=North America, disaster_type=flood — Flood depth calibration vs. human stature — operationally relevant for Pampas
D8	"Is the water level high enough to pose a drowning risk? Answer with Yes or No." disaster_type=flood, region=null — Direct life-safety relevance; generic framing not region-specific
D22	"What type of buildings are visible in the image? Select all that apply." region=North America, disaster_type=flood — Building typology options: residential houses, skyscrapers, warehouses, schools — generic North American set
D23	"What type of buildings are visible in the image? Select all that apply." region=South Asia, disaster_type=flood — Same building typology MCQ applied to South Asian flood; 'skyscrapers' and 'industrial facilities' as distractors

D30	"What type of buildings are visible in the image? Select all that apply." — choices: 'Residential houses', 'Commercial buildings', 'Skyscrapers', 'Industrial facilities' region=South Asia — Building category options do not include informal settlements (villas/asentamientos)
WEB-4	Argentine PNRRD 2025–2029 is the national disaster risk reduction plan under SINAGIR (https://www.argentina.gob.ar/sinagir/plan-2025-2029)
WEB-5	Ley N° 27.287 (SINAGIR) is the primary Argentine legislative framework (https://servicios.infoleg.gob.ar/infolegInternet/anexos/400000-404999/400480/norma.htm)
WEB-6	Decreto 225/2025 created AFE as the new national emergency authority — post-dates benchmark (https://www.boletinoficial.gob.ar/detalleAviso/primera/322808/20250321)

Strengths

- Flood is one of three most-represented disaster types with 289 images / 851 questions [Q43], and flood-specific question categories (accessibility, water level, rescue) map to core Pampas emergency manager information needs [D3, D4, D7]
- Task categories for Built Environment Damage and Access & Inaccessibility [Q50, Q51] are directly relevant to urban flood routing and damage triage
- Actionable task codes (AT-1 evacuation, AT-2 debris clearance) address the operational decisions Pampas emergency managers must support [D24]
- Hard-negative curation [Q25] tests false-positive resistance, relevant to noisy WhatsApp/social media intake

ID	Evidence
Q25	"Hard-negative images: To evaluate model resistance to false positives, we explicitly curated "hard-negative" samples, which are non-disaster images that share visual textures or chaotic elements as disaster scenes (e.g., fireworks)." (p.3)
Q43	"Earthquakes (427 images, 1,227 questions), Floods (289 images, 851 questions), and Hurricanes (265 images, 867 questions) dominate the corpus, together accounting for approximately 70% of all samples." (p.4)
Q50	"From an operational perspective, the dataset emphasizes information needs critical to humanitarian response (ref. Figure 3.c): Built Environment Damage (722 images, 1,018 questions) and Hazard Type & Severity (710 images, 912 questions) are the most prevalent categories, followed by Population/Asset Presence & Exposure (604 images, 870 questions)." (p.4)
Q51	"Access & Inaccessibility, central to routing and logistics, accounts for 410 images paired with 511 questions, while less frequent categories such as Utilities Status & Damage, Search and Rescue, and Debris Clearance & Earthmoving capture high-impact, mission-critical decision-making scenarios." (p.4)
D3	"Is this road drivable? Answer with Yes or No." disaster_type=flood, region=null — Core accessibility question directly relevant to deployment; no geographic context attached
D4	"What is the approximate water level shown on the gauge? Select all that apply." region=South Asia, disaster_type=flood — Fine-grained water level quantification; operationally relevant for Pampas flood depth assessment
D7	"What is the level of floodwater compared to the height of the people visible in the image?" region=North America, disaster_type=flood — Flood depth calibration vs. human stature — operationally relevant for Pampas
D24	"What is the primary activity visible in the image? Select all that apply." region=South Asia, disaster_type=flood, code=AT-1 — Actionable task coding — evacuation identification, relevant to Pampas use case

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: Required categories for Pampas urban flood assessment include: road passability, water depth quantification, building/infrastructure damage, evacuation activity recognition, informal-settlement flood exposure, utility damage, and refugio access. The benchmark covers most of these at the abstract level [Q50, Q51, D3, D4, D24] but lacks Argentine-specific subcategories such as informal-settlement typology and zanjón/underpass infrastructure [D22, D23, D30].

IO-2: Yes — the ontology omits Argentine-specific categories such as villa/asentamiento (informal settlement) recognition, zanjón (open stormwater drain) status, casas chorizo housing typology, and SINAGIR-defined damage classifications. Building-type MCQ options in the sample include only 'Residential houses', 'Skyscrapers', 'Warehouses', 'Schools', 'Commercial buildings', 'Industrial facilities' [D22, D23, D30] — no informal settlement category.

IO-3: Yes — the benchmark spans nine disaster categories including earthquakes, hurricanes, wildfires, landslides, accidents, and fires [Q116, Q43] that are largely irrelevant to Pampas urban flood assessment. Approximately 70% of the corpus is earthquake/flood/hurricane [Q43], meaning ~55–60% of the corpus concerns non-flood events.

IO-4: Category gaps that harm content validity: (a) absence of informal-settlement (villa/asentamiento) as a distinct built-environment class, which is a flood-vulnerable category in the Pampas [D22, D23, D30]; (b) absence of Argentine hydrological infrastructure categories (zanjones, underpasses on grid-plan streets); (c) FEMA/OCHA ontology not mapped to SINAGIR/AFE operational categories [Q9, WEB-4, WEB-6]; (d) evaluation-only framing [Q112, Q114] contradicts operational deployment intent.

ID	Evidence
Q9	"The questions are grounded in established humanitarian response frameworks (e.g., FEMA ESF (FEMA 2025), OCHA MIRA (UN OCHA 2025), etc.) and cover tasks such as identifying built environment damage, determining hazard type and severity, evaluating accessibility, and assessing movement restrictions and controls, making the dataset highly relevant for real-world disaster response applications." (p.1)
Q43	"Earthquakes (427 images, 1,227 questions), Floods (289 images, 851 questions), and Hurricanes (265 images, 867 questions) dominate the corpus, together accounting for approximately 70% of all samples." (p.4)
Q50	"From an operational perspective, the dataset emphasizes information needs critical to humanitarian response (ref. Figure 3.c): Built Environment Damage (722 images, 1,018 questions) and Hazard Type & Severity (710 images, 912 questions) are the most prevalent categories, followed by Population/Asset Presence & Exposure (604 images, 870 questions)." (p.4)
Q51	"Access & Inaccessibility, central to routing and logistics, accounts for 410 images paired with 511 questions, while less frequent categories such as Utilities Status & Damage, Search and Rescue, and Debris Clearance & Earthmoving capture high-impact, mission-critical decision-making scenarios." (p.4)
Q112	"DisasterVQA is designed as an evaluation benchmark for assessing vision–language perception and reasoning in disaster scenarios, rather than as a dataset for model training or deployment." (p.8)
Q114	"DisasterVQA is therefore best understood as a controlled diagnostic tool for identifying strengths, failure modes, and research gaps in current vision–language models, rather than as a proxy for real-world disaster response performance." (p.8)
Q116	"In this work, we have introduced DisasterVQA, a rigorously curated visual-question-answering benchmark spanning nine disaster categories and eight state-of-the-art vision–language models." (p.9)
D3	"Is this road drivable? Answer with Yes or No." disaster_type=flood, region=null — Core accessibility question directly relevant to deployment; no geographic context attached
D4	"What is the approximate water level shown on the gauge? Select all that apply." region=South Asia, disaster_type=flood — Fine-grained water level quantification; operationally relevant for Pampas flood depth assessment
D22	"What type of buildings are visible in the image? Select all that apply." region=North America, disaster_type=flood — Building typology options: residential houses, skyscrapers, warehouses, schools — generic North American set
D23	"What type of buildings are visible in the image? Select all that apply." region=South Asia, disaster_type=flood — Same building typology MCQ applied to South Asian flood; 'skyscrapers' and 'industrial facilities' as distractors
D24	"What is the primary activity visible in the image? Select all that apply." region=South Asia, disaster_type=flood, code=AT-1 — Actionable task coding — evacuation identification, relevant to Pampas use case
D30	"What type of buildings are visible in the image? Select all that apply." — choices: 'Residential houses', 'Commercial buildings', 'Skyscrapers', 'Industrial facilities' region=South Asia — Building category options do not include informal settlements (villas/asentamientos)
WEB-4	Argentine PNRRD 2025–2029 is the national disaster risk reduction plan under SINAGIR (https://www.argentina.gob.ar/sinagir/plan-2025-2029)
WEB-6	Decreto 225/2025 created AFE as the new national emergency authority — post-dates benchmark (https://www.boletinoficial.gob.ar/detalleAviso/primera/322808/20250321)

Information Gaps

- No documentation maps FEMA ESF / OCHA MIRA categories to Argentine SINAGIR/AFE operational categories
- No taxonomy for informal settlement flooding, which is a high-priority category for Pampas emergency managers

Requires Expert Verification

- Whether Argentine emergency managers would prioritize additional flood-specific subcategories (e.g., zanjón obstruction, underpass inundation, refugio access) not currently in the taxonomy
- Whether SINAGIR/AFE damage classification thresholds can be reconciled with FEMA ESF categories

Remediation

Gap 1: Ontology omits villa/asentamiento recognition, zanjón/underpass status, and refugio evacuation-route viability — categories central to Pampas emergency operations

Recommendation: Add these categories to the task taxonomy via a Pampas-extension subset, developed jointly with SINAGIR/AFE liaisons and municipal civil protection representatives from Greater Buenos Aires, Rosario, Santa Fe, and Córdoba.

Input Content (IC)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: DISASTERVQA | **Context:** Pampas Region Emergency Managers (Argentina) — Urban Flood Damage Assessment

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input content is fundamentally misaligned with the target deployment context. The paper explicitly acknowledges geographic skew toward South Asia and North America [Q48] and inherited social-media biases [Q107, Q108]. Empirical inspection confirms that of 254 sampled examples, only 5 carry a 'South America' region tag — and all 5 are hurricane, not flood; furthermore, at least one South America-tagged image has 'Auto-rickshaw' as a correct answer option [D13], suggesting geographic misattribution. Pampas-specific visual signatures — grid-plan streets, casas chorizo, conventillos, villas/asentamientos, zanjones — are not present in the accessible sample. Flood imagery in the sample overwhelmingly reflects South Asian contexts: boats, wooden log crossings, metal-container rescues, motorcycles submerged [D1, D14, D26, D28]. The Pampas has hydro-geomorphological features (flat plain, clay soils, rising groundwater tables) that differ substantially from South Asian and North American flood contexts [WEB-1, WEB-2]. No Pampas-specific content curation or verification was performed.

ID	Evidence
Q48	"The dataset spans a broad set of regions (as shown in Figure 3.b), with strong representation from South Asia and North America, reflecting population density and reporting bias in social media-derived data." (p.4)
Q107	"Because the dataset is constructed by sampling images from existing publicly available sources, it inevitably inherits their underlying biases, despite our careful curation and expert verification procedures aimed at mitigating these effects." (p.8)
Q108	"In particular, social media-sourced imagery exhibits geographic, demographic, temporal, and device-related biases, with disproportionate representation of urban, well-connected regions and visually salient events." (p.8)
D1	"Is a boat being used for rescue or transport? Answer with Yes or No." region=South Asia, disaster_type=flood — Flood rescue question; ground-truth affirms boat use in South Asian context
D13	"What types of vehicles are visible in the image? Select all that apply." region=South America, disaster_type=hurricane — Auto-rickshaw as an answer option signals non-Argentine infrastructure context
D14	"What objects are being used to create a makeshift crossing? Select all that apply." region=South Asia, disaster_type=flood — Improvised flood crossing — South Asian context, not Pampas urban setting
D26	"What objects are submerged in the floodwaters? Select all that apply." region=South Asia, disaster_type=flood — Objects submerged in South Asian floodwater; motorcycle consistent with regional transport
D28	"What object is being used to transport the person in the image?" region=South Asia, disaster_type=flood — Improvised flood transport — metal container; reflects South Asian rescue practice
WEB-1	Pampas is a flat plain watershed (mean slope ~1 m/km) with clay-heavy soils of low permeability — hydro-geomorphology differs substantially from South Asian/North American flood contexts (https://pmc.ncbi.nlm.nih.gov/articles/PMC7417906/)
WEB-2	2023 CONICET/Science study documented rising Pampas groundwater tables driven by soy monoculture, producing a lower-intensity peri-urban inundation visual regime not represented in DisasterVQA (https://puntoconvergente.uca.edu.ar/despues-de-las-inundaciones-bahia-blanca-que-se-necesita-para-evitar-otra-tragedia/)

Strengths

- All source imagery is derived from social media / crowdsourced repositories (CrisisMMD, Incidents1M, MEDIC) [Q20, D1, D26], consistent with the deployment channel (WhatsApp/Twitter/Instagram imagery from affected populations)

- Flood is one of three dominant disaster categories with 289 images and 851 questions [Q43], providing substantive volume for flood-specific evaluation even if geographically skewed
- Ground-level (not aerial) imagery [Q104], which matches the perspective of population-submitted photos in the deployment scenario

ID	Evidence
Q20	"We sourced our base imagery from three established datasets: CrisisMMD (Alam, Ofli, and Imran 2018), Incidents1M (Weber et al. 2023), and MEDIC (Alam et al. 2023)." (p.3)
Q43	"Earthquakes (427 images, 1,227 questions), Floods (289 images, 851 questions), and Hurricanes (265 images, 867 questions) dominate the corpus, together accounting for approximately 70% of all samples." (p.4)
Q104	"Because disaster imagery is inherently scarce and complex, we prioritized expert-curated, high-quality annotations strictly grounded in humanitarian frameworks over massive, noisy data scraping." (p.8)
D1	"Is a boat being used for rescue or transport? Answer with Yes or No." region=South Asia, disaster_type=flood — Flood rescue question; ground-truth affirms boat use in South Asian context
D26	"What objects are submerged in the floodwaters? Select all that apply." region=South Asia, disaster_type=flood — Objects submerged in South Asian floodwater; motorcycle consistent with regional transport

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes — Pampas flood assessment requires recognition of Argentine urban infrastructure (grid-plan streets, casas chorizo, conventillos, villas/asentamientos, zanjones) and Pampas hydro-geomorphology (flat plain, clay soils, rising groundwater tables driven by soy monoculture) [WEB-1, WEB-2]. None of these regionally specific visual signatures are documented as represented in the benchmark [Q48].

IC-2: Culturally sensitive content is not aligned. Empirical inspection found only 3 unique 'South America'-tagged images among 254 samples, all hurricane, with at least one likely misattributed (auto-rickshaw answer option) [D13]. Argentine flood visual signatures are effectively absent from the sample.

IC-3: Yes — flood imagery in the sample reflects South Asian rescue practices (metal containers, wooden log crossings, boats) [D14, D28, D1] and North American infrastructure (hurricane damage patterns) [D19] that would not transfer to Pampas urban flood contexts.

IC-4: No regional annotators from Argentina or Latin America were involved. Documentation shows only Qatar-based institutional affiliations [Q11], and no Latin American expert validation is documented.

IC-5: Content validity is substantially harmed: (a) geographic skew to South Asia / North America [Q48]; (b) inherited social-media biases toward dramatic events and well-connected urban regions [Q108, Q109]; (c) empirical absence of Pampas-representative flood imagery [D13, dataset summary]; (d) 682 images with no geolocation at all [Q49, D3, D8]; (e) 2023 CONICET/Science study documents rising Pampas groundwater causing lower-intensity peri-urban inundation, a visual regime even further from DisasterVQA distribution [WEB-2].

ID	Evidence
Q11	"Qatar Computing Research Institute, Hamad Bin Khalifa University, Doha, Qatar; College of Science & Engineering, Hamad Bin Khalifa University, Doha, Qatar; Department of Computer Science & Engineering, Qatar University, Doha, Qatar" (p.1)

Q48	"The dataset spans a broad set of regions (as shown in Figure 3.b), with strong representation from South Asia and North America, reflecting population density and reporting bias in social media–derived data." (p.4)
Q49	"A substantial portion of samples (682 images with 2,279 questions), primarily sourced from Incident1M, are labeled as N/A due to missing or ambiguous location information." (p.4)
Q108	"In particular, social media–sourced imagery exhibits geographic, demographic, temporal, and device-related biases, with disproportionate representation of urban, well-connected regions and visually salient events." (p.8)
Q109	"Platform-level amplification further skews coverage toward dramatic scenes, while subtler yet operationally important impacts may be underrepresented." (p.8)
D1	"Is a boat being used for rescue or transport? Answer with Yes or No." region=South Asia, disaster_type=flood — Flood rescue question; ground-truth affirms boat use in South Asian context
D3	"Is this road drivable? Answer with Yes or No." disaster_type=flood, region=null — Core accessibility question directly relevant to deployment; no geographic context attached
D8	"Is the water level high enough to pose a drowning risk? Answer with Yes or No." disaster_type=flood, region=null — Direct life-safety relevance; generic framing not region-specific
D13	"What types of vehicles are visible in the image? Select all that apply." region=South America, disaster_type=hurricane — Auto-rickshaw as an answer option signals non-Argentine infrastructure context
D14	"What objects are being used to create a makeshift crossing? Select all that apply." region=South Asia, disaster_type=flood — Improvised flood crossing — South Asian context, not Pampas urban setting
D19	"Which objects are visibly damaged in the image?" region=North America, disaster_type=hurricane — Utility infrastructure damage MCQ; North America context
D28	"What object is being used to transport the person in the image?" region=South Asia, disaster_type=flood — Improvised flood transport — metal container; reflects South Asian rescue practice
WEB-1	Pampas is a flat plain watershed (mean slope ~1 m/km) with clay-heavy soils of low permeability — hydro-geomorphology differs substantially from South Asian/North American flood contexts (https://pmc.ncbi.nlm.nih.gov/articles/PMC7417906/)
WEB-2	2023 CONICET/Science study documented rising Pampas groundwater tables driven by soy monoculture, producing a lower-intensity peri-urban inundation visual regime not represented in DisasterVQA (https://puntoconvergente.uca.edu.ar/despues-de-las-inundaciones-bahia-blanca-que-se-necesita-para-evitar-otra-tragedia/)

Information Gaps

- Full geographic distribution of the 682 null-region images is unknown; a small Argentine subset could theoretically exist but is not identifiable from metadata
- Actual image content (not inspectable in this analysis) could contain incidental Pampas features, though the region-tag evidence suggests this is unlikely at meaningful volume

Requires Expert Verification

- Whether the small number of South America–tagged flood images (if any exist outside the 254-example sample) contain Argentine content
- Direct visual inspection by an Argentine emergency manager to identify whether any imagery matches Pampas visual signatures

Remediation

Gap 1: Absence of Pampas / Argentine flood imagery; South America tag in the dataset does not correspond to Argentine flood content (only 5 SA-tagged items in 254 sampled, all hurricane, at least one likely misattributed)

Recommendation: Curate a supplementary Argentine flood evaluation set of 200–400 ground-level, social-media-derived images from the 2013 La Plata, 2015 RMBA, 2023 and 2025 Bahía Blanca, and 2025 Pampas events, geotagged and annotated by Argentine emergency management stakeholders. Use as a targeted validation overlay rather than a replacement.

Input Form (IF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DISASTERVQA | **Context:** Pampas Region Emergency Managers (Argentina) — Urban Flood Damage Assessment

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Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark uses static images only [\[Q104, dataset metadata\]](#), while the deployment use case explicitly includes video (WhatsApp video clips, TikTok, Instagram Stories, as documented in the regional YAML). All prompts and input questions are in English [\[Q1 examples throughout registry; dataset schema\]](#), while Pampas emergency managers operate in Rioplatense Spanish. The image selection prioritized high visual complexity and multi-object scenes [\[Q23\]](#), and low-signal images were discarded [\[Q24\]](#), which may exclude the subtler, lower-quality imagery typical of WhatsApp forwards under adverse network conditions. Argentine urban connectivity supports the technical capture parameters (89.67% internet, ~90% mobile, ~98% smartphone ownership among internet users) [\[WEB-12, WEB-13, WEB-15\]](#), so infrastructure is not the limiting factor — the mismatch is in modality (image-only vs. image+video) and language (English-only vs. Spanish).

ID	Evidence
Q1	"DisasterVQA consists of 1,395 real-world images and 4,405 expert-curated question–answer pairs spanning diverse events such as floods, wildfires, and earthquakes." (p.1)
Q23	"The selection strategy prioritized semantic density over volume, adhering to two strict criteria: Disaster-positive images: Annotators selected only those images exhibiting high visual complexity and multi-object scenes clearly indicative of disaster events (e.g., floods, wildfires, earthquakes)." (p.3)
Q24	"Images with low visual signal or minimal reasoning value were discarded to ensure every positive sample posed a genuine interpretative challenge." (p.3)
Q104	"Because disaster imagery is inherently scarce and complex, we prioritized expert-curated, high-quality annotations strictly grounded in humanitarian frameworks over massive, noisy data scraping." (p.8)
WEB-12	Argentine internet penetration 89.67 per 100 people (2024) (https://fred.stlouisfed.org/series/ITNETUSERP2ARG)
WEB-13	Mobile internet penetration ~90% of population in 2024 (https://www.statista.com/statistics/244986/mobile-internet-user-penetration-in-argentina-since-2010/)
WEB-15	Smartphone ownership >98% among Argentine internet users (2021) (https://www.emarketer.com/content/global-media-intelligence-2021-argentina)

Strengths

- Image-based social-media-sourced content is consistent with a significant portion of the deployment input stream [\[Q20, D1\]](#)
- FAIR-principle data formatting with JSON schemas [\[Q125\]](#) supports integration into Argentine EOC pipelines
- Ground-level imagery matches the perspective of population-submitted photos
- Nighttime/low-light imagery is present in at least some samples [\[D31\]](#), covering an operationally relevant capture condition

ID	Evidence
Q20	"We sourced our base imagery from three established datasets: CrisisMMD (Alam, Ofli, and Imran 2018), Incidents1M (Weber et al. 2023), and MEDIC (Alam et al. 2023)." (p.3)

Q125	"We designed DisasterVQA with the FAIR principles in mind. The dataset will be Findable through a public repository with persistent identifiers and detailed metadata; Accessible via an open license and standard download mechanisms; Interoperable by using widely adopted data formats (e.g., JSON for image-question-answer triplets) and clearly defined schemas; and Reusable through comprehensive documentation, data curation guidelines, and evaluation protocols that enable reproducibility and downstream use." (p.11)
D1	"Is a boat being used for rescue or transport? Answer with Yes or No." region=South Asia, disaster_type=flood — Flood rescue question; ground-truth affirms boat use in South Asian context
D31	"What time of day is depicted in the image? Select all that apply." region=Southeast Asia, disaster_type=hurricane — Temporal context question — night imagery; low-light condition relevant to deployment

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Static image resolution and encoding are compatible with Argentine device infrastructure (~98% smartphone ownership among internet users [WEB-15]; 4G-dominant urban connectivity [WEB-16]). No documented mismatch in image signal parameters. However, image-only modality omits video, which is a primary channel in the deployment context.

IF-2: Yes — Argentine urban infrastructure supports the same capture specifications; connectivity is adequate for image/video ingestion in urban Pampas [WEB-12, WEB-13, WEB-16]. Rural/peri-urban and EOC-specific connectivity remains unverified [regional YAML: infrastructure_notes].

IF-3: Two significant domain-specific form differences: (a) video modality is absent from the benchmark [Q104, dataset schema showing only image/text modalities] but explicitly part of the deployment input stream (WhatsApp video clips, TikTok, Instagram Stories); (b) prompt language is English-only [dataset schema, all examples in Verbatim Quote Registry] while operational Spanish (Rioplátense) is required.

IF-4: Form mismatches that harm external validity: (1) no video evaluation — benchmark cannot predict model behavior on video frames or motion-blur imagery; (2) no Spanish-language prompting — cross-lingual prompt sensitivity in VLMs means English scores cannot be assumed to predict Spanish performance; (3) selection bias toward high-complexity images [Q23, Q24] may under-represent the low-signal WhatsApp content typical of the deployment.

ID	Evidence
Q23	"The selection strategy prioritized semantic density over volume, adhering to two strict criteria: Disaster-positive images: Annotators selected only those images exhibiting high visual complexity and multi-object scenes clearly indicative of disaster events (e.g., floods, wildfires, earthquakes)." (p.3)
Q24	"Images with low visual signal or minimal reasoning value were discarded to ensure every positive sample posed a genuine interpretative challenge." (p.3)
Q104	"Because disaster imagery is inherently scarce and complex, we prioritized expert-curated, high-quality annotations strictly grounded in humanitarian frameworks over massive, noisy data scraping." (p.8)
WEB-12	Argentine internet penetration 89.67 per 100 people (2024) (https://fred.stlouisfed.org/series/ITNETUSERP2ARG)
WEB-13	Mobile internet penetration ~90% of population in 2024 (https://www.statista.com/statistics/244986/mobile-internet-user-penetration-in-argentina-since-2010/)
WEB-15	Smartphone ownership >98% among Argentine internet users (2021) (https://www.emarketer.com/content/global-media-intelligence-2021-argentina)
WEB-16	4G mobile dominant in urban Pampas; average mobile download speed 62.64 Mbit/s (https://www.worlddata.info/america/argentina/telecommunication.php)

Information Gaps

- EOC-specific connectivity in the Pampas is not documented; may constrain real-time video ingestion
- The extent to which VLMs benchmarked on English DisasterVQA prompts would degrade under Spanish prompting is not empirically established for these specific models

Requires Expert Verification

- Actual WhatsApp video volume vs. still-image volume in Pampas emergency intake — an ethnographic/operational question
-

Remediation

Gap 1: English-only prompts and image-only modality; deployment requires Rioplatense Spanish and includes video input

Recommendation: Create parallel Spanish-language (Rioplatense) versions of the flood-subset QA pairs and add a WhatsApp-representative video subset (10–30 second clips) with frame-level and clip-level questions. Report separate metrics for English vs. Spanish prompting and image vs. video inputs to quantify cross-modal and cross-lingual sensitivity.

Output Ontology (OO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: DISASTERVQA | **Context:** Pampas Region Emergency Managers (Argentina) — Urban Flood Damage Assessment

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output categories are grounded in FEMA ESF and OCHA MIRA frameworks [Q38, Q50, Q51], which are internationally oriented but not calibrated to Argentine SINAGIR/AFE structures [WEB-4, WEB-5, WEB-6]. The three output formats (binary, MCQ, open-ended) [Q8, Q37] are structurally appropriate and provide useful diagnostic granularity. Some output categories map well to Pampas emergency manager needs — Access & Inaccessibility, Built Environment Damage, Water & Flood Characteristics [Q50, Q51, D5, D11] — and empirical inspection confirms graduated flood depth options ('below the knees', 'up to the waist', 'up to the roof') that are operationally interpretable [D7, D11, D15]. However, MCQ option sets omit Argentine-relevant categories: building typology options do not include informal settlements (villas/asentamientos) [D22, D23, D30]; cautionary object vocabulary reflects Anglo-American conventions [D20]; crisis_info_codes reference FEMA ESF, not SINAGIR classifications [D24, D5, D8]. Argentina aligns with Sendai Framework at the strategic level [WEB-5, WEB-4] but operational terminology differs from FEMA/OCHA. Decision thresholds for damage severity have not been validated against Argentine standards.

ID	Evidence
Q8	"The questions are designed across three formats: binary, multiple-choice, and open-ended, to comprehensively assess VLMs' reasoning capabilities under varied levels of difficulty and ambiguity." (p.1)
Q37	"The final human-verified dataset contains 4,405 question–answer pairs, including 2,153 binary (yes/no), 1,693 multiple-choice, and 559 open-ended questions." (p.3)
Q38	"We employed GPT-4o to categorize each verified Q&A pair based on definitions from several frameworks, including the FEMA Emergency Support Functions (ESF) (FEMA 2025) and OCHA MIRA (UN OCHA 2025)." (p.3)
Q50	"From an operational perspective, the dataset emphasizes information needs critical to humanitarian response (ref. Figure 3.c): Built Environment Damage (722 images, 1,018 questions) and Hazard Type & Severity (710 images, 912 questions) are the most prevalent categories, followed by Population/Asset Presence & Exposure (604 images, 870 questions)." (p.4)
Q51	"Access & Inaccessibility, central to routing and logistics, accounts for 410 images paired with 511 questions, while less frequent categories such as Utilities Status & Damage, Search and Rescue, and Debris Clearance & Earthmoving capture high-impact, mission-critical decision-making scenarios." (p.4)
D5	"What is the condition of the road in the image? Select all that apply." region=South Asia, disaster_type=flood — Road accessibility grading relevant to routing decisions
D7	"What is the level of floodwater compared to the height of the people visible in the image?" region=North America, disaster_type=flood — Flood depth calibration vs. human stature — operationally relevant for Pampas
D8	"Is the water level high enough to pose a drowning risk? Answer with Yes or No." disaster_type=flood, region=null — Direct life-safety relevance; generic framing not region-specific
D11	"What is the level of floodwater compared to the height of the submerged vehicle?" region=South Asia, disaster_type=flood — Detailed water depth MCQ relevant to Pampas flood assessment
D15	"What is the level of floodwater compared to the height of the people visible in the image?" disaster_type=flood, region=null — Flood depth relative to humans — valid for deployment; no regional grounding
D20	"What is the visible cautionary object? Answer with one or two words." disaster_type=landslide — Cautionary signage groundtruth uses North American/generic terminology
D22	"What type of buildings are visible in the image? Select all that apply." region=North America, disaster_type=flood — Building typology options: residential houses, skyscrapers, warehouses, schools — generic North American set

D23	"What type of buildings are visible in the image? Select all that apply." region=South Asia, disaster_type=flood — Same building typology MCQ applied to South Asian flood; 'skyscrapers' and 'industrial facilities' as distractors
D24	"What is the primary activity visible in the image? Select all that apply." region=South Asia, disaster_type=flood, code=AT-1 — Actionable task coding — evacuation identification, relevant to Pampas use case
D30	"What type of buildings are visible in the image? Select all that apply." — choices: 'Residential houses', 'Commercial buildings', 'Skyscrapers', 'Industrial facilities' region=South Asia — Building category options do not include informal settlements (villas/asentamientos)
WEB-4	Argentine PNRRD 2025–2029 is the national disaster risk reduction plan under SINAGIR (https://www.argentina.gob.ar/sinagir/plan-2025-2029)
WEB-5	Ley N° 27.287 (SINAGIR) is the primary Argentine legislative framework (https://servicios.infoleg.gob.ar/infolegInternet/anexos/400000-404999/400480/norma.htm)
WEB-6	Decreto 225/2025 created AFE as the new national emergency authority — post-dates benchmark (https://www.boletinoficial.gob.ar/detalleAviso/primera/322808/20250321)

Strengths

- Graduated water-depth output categories (knees/waist/chest/roof) are operationally interpretable across contexts [D7, D11, D15]
- Access & Inaccessibility category with binary, MCQ, and open-ended framing directly maps to Pampas routing decisions [Q51, D3, D5]
- Multi-label MCQ format captures ambiguity of real disaster scenes [D19], appropriate for messy Pampas flood imagery
- AT/SA coding distinguishes actionable from situational-awareness outputs [Q39, D24], which aligns with the deployment need to prioritize actionable decisions
- Sendai Framework alignment provides strategic-level compatibility between international humanitarian categories and Argentine national plan (PNRRD 2025–2029) [WEB-4, WEB-5]

ID	Evidence
Q39	"We instructed the model to classify questions into two operational themes: Situational Awareness and Actionable Information." (p.3)
Q51	"Access & Inaccessibility, central to routing and logistics, accounts for 410 images paired with 511 questions, while less frequent categories such as Utilities Status & Damage, Search and Rescue, and Debris Clearance & Earthmoving capture high-impact, mission-critical decision-making scenarios." (p.4)
D3	"Is this road drivable? Answer with Yes or No." disaster_type=flood, region=null — Core accessibility question directly relevant to deployment; no geographic context attached
D5	"What is the condition of the road in the image? Select all that apply." region=South Asia, disaster_type=flood — Road accessibility grading relevant to routing decisions
D7	"What is the level of floodwater compared to the height of the people visible in the image?" region=North America, disaster_type=flood — Flood depth calibration vs. human stature — operationally relevant for Pampas
D11	"What is the level of floodwater compared to the height of the submerged vehicle?" region=South Asia, disaster_type=flood — Detailed water depth MCQ relevant to Pampas flood assessment
D15	"What is the level of floodwater compared to the height of the people visible in the image?" disaster_type=flood, region=null — Flood depth relative to humans — valid for deployment; no regional grounding
D19	"Which objects are visibly damaged in the image?" region=North America, disaster_type=hurricane — Utility infrastructure damage MCQ; North America context
D24	"What is the primary activity visible in the image? Select all that apply." region=South Asia, disaster_type=flood, code=AT-1 — Actionable task coding — evacuation identification, relevant to Pampas use case
WEB-4	Argentine PNRRD 2025–2029 is the national disaster risk reduction plan under SINAGIR (https://www.argentina.gob.ar/sinagir/plan-2025-2029)
WEB-5	Ley N° 27.287 (SINAGIR) is the primary Argentine legislative framework (https://servicios.infoleg.gob.ar/infolegInternet/anexos/400000-404999/400480/norma.htm)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: Output labels are partially relevant: water depth grading, road passability, damage severity, and evacuation identification map to Pampas emergency needs [D5, D7, D11, D24]. However, FEMA ESF classification of activities like 'Firefighting', 'Medical aid', 'Search and rescue' [D24] uses ESF vocabulary that Argentine SINAGIR/AFE managers would not use in operational reporting.

OO-2: Missing regional categories: (a) informal settlement (villa/asentamiento) as a building-type MCQ option [D22, D23, D30]; (b) Argentine-specific damage severity thresholds mapped to SINAGIR damage classifications; (c) Rioplatense Spanish-language output categories; (d) refugio (evacuation shelter) recognition as an actionable-task subcategory.

OO-3: Yes — FEMA ESF and OCHA MIRA encode Anglo-American / UN humanitarian assumptions [Q9, Q38] that do not map directly to Argentine institutional frameworks (SINAGIR under Ley 27.287; AFE under Decreto 225/2025) [WEB-4, WEB-5, WEB-6]. Cautionary object vocabulary in ground truth (caution tape / barrier tape / police tape) reflects Anglo-American signage conventions [D20].

OO-4: Stakeholder-driven taxonomy redesign would be needed for operational deployment: mapping FEMA ESF categories to SINAGIR damage classifications, adding Argentine-specific building typologies, and providing Spanish-language equivalents. This has not been performed.

OO-5: Structural and content validity concerns: (a) FEMA/OCHA structure does not mirror SINAGIR operational structure, harming structural validity; (b) omitted informal-settlement category harms content validity in a region where villa flooding is a high-priority concern; (c) English-only categories harm external validity for Spanish-speaking users. However, categories are broadly plausible and not directly harmful — hence a middle score rather than lower.

ID	Evidence
Q9	"The questions are grounded in established humanitarian response frameworks (e.g., FEMA ESF (FEMA 2025), OCHA MIRA (UN OCHA 2025), etc.) and cover tasks such as identifying built environment damage, determining hazard type and severity, evaluating accessibility, and assessing movement restrictions and controls, making the dataset highly relevant for real-world disaster response applications." (p.1)
Q38	"We employed GPT-4o to categorize each verified Q&A pair based on definitions from several frameworks, including the FEMA Emergency Support Functions (ESF) (FEMA 2025) and OCHA MIRA (UN OCHA 2025)." (p.3)
D5	"What is the condition of the road in the image? Select all that apply." region=South Asia, disaster_type=flood — Road accessibility grading relevant to routing decisions
D7	"What is the level of floodwater compared to the height of the people visible in the image?" region=North America, disaster_type=flood — Flood depth calibration vs. human stature — operationally relevant for Pampas
D11	"What is the level of floodwater compared to the height of the submerged vehicle?" region=South Asia, disaster_type=flood — Detailed water depth MCQ relevant to Pampas flood assessment
D20	"What is the visible cautionary object? Answer with one or two words." disaster_type=landslide — Cautionary signage groundtruth uses North American/generic terminology
D22	"What type of buildings are visible in the image? Select all that apply." region=North America, disaster_type=flood — Building typology options: residential houses, skyscrapers, warehouses, schools — generic North American set
D23	"What type of buildings are visible in the image? Select all that apply." region=South Asia, disaster_type=flood — Same building typology MCQ applied to South Asian flood; 'skyscrapers' and 'industrial facilities' as distractors
D24	"What is the primary activity visible in the image? Select all that apply." region=South Asia, disaster_type=flood, code=AT-1 — Actionable task coding — evacuation identification, relevant to Pampas use case

D30	"What type of buildings are visible in the image? Select all that apply." — choices: 'Residential houses', 'Commercial buildings', 'Skyscrapers', 'Industrial facilities' region=South Asia — Building category options do not include informal settlements (villas/asentamientos)
WEB-4	Argentine PNRRD 2025–2029 is the national disaster risk reduction plan under SINAGIR (https://www.argentina.gob.ar/sinagir/plan-2025-2029)
WEB-5	Ley N° 27.287 (SINAGIR) is the primary Argentine legislative framework (https://servicios.infoleg.gob.ar/infolegInternet/anexos/400000-404999/400480/norma.htm)
WEB-6	Decreto 225/2025 created AFE as the new national emergency authority — post-dates benchmark (https://www.boletinoficial.gob.ar/detalleAviso/primera/322808/20250321)

Information Gaps

- Explicit mapping between FEMA ESF categories and SINAGIR damage/response classifications does not exist in the benchmark or the regional YAML
- Whether specific SINAGIR reporting fields have direct FEMA analogs is undocumented

Requires Expert Verification

- Whether Argentine emergency managers accept FEMA ESF-derived categories as operationally meaningful in their reporting chains
- Whether the graduated water-depth categories match SINAGIR/AFE flood severity thresholds

Remediation

Gap 1: FEMA ESF / OCHA MIRA categories not mapped to Argentine SINAGIR/AFE operational frameworks, and MCQ option sets omit informal-settlement (villa/asentamiento) building typology

Recommendation: Develop an explicit mapping between FEMA ESF categories and SINAGIR damage/response classifications (validated by AFE technical staff), and extend MCQ option sets to include villa/asentamiento, casas chorizo, conventillo, and Argentine hydrological infrastructure (zanjón, underpass) categories.

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DISASTERVQA | Context: Pampas Region Emergency Managers (Argentina) — Urban Flood Damage Assessment

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Ground-truth labels were produced by a multi-stage pipeline [Q19] with Group A (two annotators) reviewing GPT-4o-generated candidates [Q32, Q33] and Group B (two senior researchers with a decade+ of experience in AI for disaster response) performing final QA [Q34, Q35]. However, all annotators are affiliated with Qatar-based institutions [Q11], and no Latin American, Argentine, or Rioplatense Spanish-speaking regional expertise is documented. The GPT-4o LLM-judge used for open-ended evaluation was validated against a stratified sample of 196 model-image pairs achieving 99.5% accuracy [Q69, Q70] — but this validation itself was performed by the same annotator pool, providing no cross-cultural check. Empirical inspection confirms label content reflects Anglo-American conventions (caution tape / barrier tape / police tape terminology [D20]) and South Asian rescue practices (metal container transport [D28], wooden log crossings [D14]). Convergent validity with Argentine stakeholder judgments has not been assessed. The absence of Argentine annotators combined with the Anglo/South-Asian visual and lexical content produces substantive concerns about label correctness for a Pampas deployment context.

ID	Evidence
Q11	"Qatar Computing Research Institute, Hamad Bin Khalifa University, Doha, Qatar; College of Science & Engineering, Hamad Bin Khalifa University, Doha, Qatar; Department of Computer Science & Engineering, Qatar University, Doha, Qatar" (p.1)
Q19	"To achieve these goals, we employed a multi-stage, human-in-the-loop pipeline (as shown in Figure 2) that integrates diverse source imagery with expert-verified annotations, as detailed in the following subsections." (p.2)
Q32	"Group A is a team of two annotators who reviewed the raw GPT-4o output for triviality and correctness." (p.3)
Q33	"They also corrected the model's incorrect answers and added new questions to address identified gaps." (p.3)
Q34	"Group B, consisting of two senior researchers with over a decade of experience in AI for disaster response, performed the final quality assurance pass." (p.3)
Q35	"This group focused on semantic consistency and domain specificity, retaining questions that posed meaningful challenges for humanitarian analysis." (p.3)
Q69	"Furthermore, to validate the reliability of the GPT-4o judge in evaluating open-ended responses, a human-in-the-loop verification was conducted across a stratified sample of 196 model-image pairs (28 unique samples for each of the seven subject models)." (p.5)
Q70	"The expert audit revealed an overall judge accuracy of 99.5%, demonstrating that the post-processing extraction logic and semantic alignment are highly robust." (p.5)
D14	"What objects are being used to create a makeshift crossing? Select all that apply." region=South Asia, disaster_type=flood — Improvised flood crossing — South Asian context, not Pampas urban setting
D20	"What is the visible cautionary object? Answer with one or two words." disaster_type=landslide — Cautionary signage groundtruth uses North American/generic terminology
D28	"What object is being used to transport the person in the image?" region=South Asia, disaster_type=flood — Improvised flood transport — metal container; reflects South Asian rescue practice

Strengths

- Multi-stage human-in-the-loop pipeline with two review layers [Q19, Q32, Q34] provides quality control at the internal-consistency level

- Group B annotators are senior researchers with >10 years of AI-for-disaster-response experience [Q34], giving domain expertise (though not regional expertise)
- GPT-4o judge validated to 99.5% accuracy against expert audit [Q69, Q70] — provides confidence that the evaluation pipeline reliably applies the labels as defined
- Datasheet provided [Q126, Q115] supports transparency about known limitations

ID	Evidence
Q19	"To achieve these goals, we employed a multi-stage, human-in-the-loop pipeline (as shown in Figure 2) that integrates diverse source imagery with expert-verified annotations, as detailed in the following subsections." (p.2)
Q32	"Group A is a team of two annotators who reviewed the raw GPT-4o output for triviality and correctness." (p.3)
Q34	"Group B, consisting of two senior researchers with over a decade of experience in AI for disaster response, performed the final quality assurance pass." (p.3)
Q69	"Furthermore, to validate the reliability of the GPT-4o judge in evaluating open-ended responses, a human-in-the-loop verification was conducted across a stratified sample of 196 model-image pairs (28 unique samples for each of the seven subject models)." (p.5)
Q70	"The expert audit revealed an overall judge accuracy of 99.5%, demonstrating that the post-processing extraction logic and semantic alignment are highly robust." (p.5)
Q115	"To support responsible and transparent research use, DisasterVQA is released in alignment with the FAIR principles (FORCE11 2020) and accompanied by a Datasheet (Geburu et al. 2021)." (p.8)
Q126	"A formal Datasheet for Datasets is provided in the official repository to document the collection, preprocessing, and ethical considerations of the DisasterVQA benchmark." (p.11)

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: No — ground truth labels do not reflect Argentine stakeholder perspectives. All annotators are Qatar-based [Q11], and no Latin American or Argentine expert validation is documented. Label content includes Anglo-American vocabulary [D20] and South Asian rescue-practice interpretations [D14, D28] that would not match Argentine emergency manager judgments.

OC-2: Disagreement is likely. Argentine emergency managers might: (a) classify villas/asentamientos as a distinct building category rather than 'Residential houses'; (b) apply different damage severity thresholds calibrated to SINAGIR reporting requirements; (c) use different terminology (e.g., 'zanjón obstruido' vs. 'blocked drainage ditch') that would fail semantic-equivalence checks by an English-language LLM judge.

OC-3: Annotator demographics are only partially documented. Institutional affiliations (QCRI, HBKU, Qatar University) are stated [Q11]; individual annotator nationalities, first languages, and regional expertise are not disclosed. No datasheet section documents Latin American or Argentine annotators.

OC-4: Label re-annotation by an Argentine emergency manager panel has not been performed and would be required for deployment validity.

OC-5: Aggregation methods are not fully documented; open-ended answers use GPT-4o LLM-as-judge [Q67, Q68], whose training distribution favors English and dominant global perspectives — likely to erase minority (including Argentine) linguistic and semantic framings.

OC-6: Label validity harms: (a) convergent validity is unassessed for Argentine annotators [Q11, no documented Latin American reviewers]; (b) external validity is limited — English-language, Qatar-annotated labels may not generalize to Rioplatense Spanish operational context; (c) LLM judge is English-only [Q67] with no documented Spanish calibration.

ID	Evidence
Q11	"Qatar Computing Research Institute, Hamad Bin Khalifa University, Doha, Qatar; College of Science & Engineering, Hamad Bin Khalifa University, Doha, Qatar; Department of Computer Science & Engineering, Qatar University, Doha, Qatar" (p.1)
Q32	"Group A is a team of two annotators who reviewed the raw GPT-4o output for triviality and correctness." (p.3)
Q34	"Group B, consisting of two senior researchers with over a decade of experience in AI for disaster response, performed the final quality assurance pass." (p.3)
Q67	"For open-ended questions, we adopt an LLM-as-a-judge approach with GPT-4o to assess answer correctness in addition to output normalization." (p.5)
Q68	"The judge model compares the extracted response against the ground-truth answer, accounting for semantic equivalence, paraphrasing, and formatting variations, and produces a structured decision (Correct or Incorrect) along with the normalized answer." (p.5)
D14	"What objects are being used to create a makeshift crossing? Select all that apply." region=South Asia, disaster_type=flood — Improvised flood crossing — South Asian context, not Pampas urban setting
D20	"What is the visible cautionary object? Answer with one or two words." disaster_type=landslide — Cautionary signage groundtruth uses North American/generic terminology
D22	"What type of buildings are visible in the image? Select all that apply." region=North America, disaster_type=flood — Building typology options: residential houses, skyscrapers, warehouses, schools — generic North American set
D28	"What object is being used to transport the person in the image?" region=South Asia, disaster_type=flood — Improvised flood transport — metal container; reflects South Asian rescue practice

Information Gaps

- Individual annotator nationalities, first languages, and regional expertise are not disclosed in the paper
- No inter-annotator agreement statistics reported for Argentine or Latin American reviewers (because none were involved)
- Whether the GPT-4o LLM judge's semantic-equivalence checks maintain accuracy for Spanish-language paraphrases is unknown

Requires Expert Verification

- Re-annotation of a stratified sample by Argentine emergency managers to measure inter-cultural label agreement rates
- Direct assessment of GPT-4o judge accuracy when evaluating Spanish-language open-ended responses

Remediation

Gap 1: No Argentine or Latin American annotator involvement; label validity for the Pampas context is unassessed and English-only LLM judge is uncalibrated for Spanish semantics

Recommendation: Conduct a re-annotation study on a stratified sample of ~200 flood-relevant items by 3–5 Argentine emergency managers from municipal/provincial civil protection bodies. Report inter-cultural label agreement rates. Additionally, validate GPT-4o (or replacement) LLM-judge accuracy on Spanish-language open-ended responses via a bilingual expert audit.

Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: DISASTERVQA | **Context:** Pampas Region Emergency Managers (Argentina) — Urban Flood Damage Assessment

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The benchmark evaluates three metric types — binary accuracy, MCQ F1, open-ended accuracy via LLM-judge [Q74–Q77] — with deterministic settings [Q54, Q123] and structured output normalization [Q65, Q66]. All outputs are English text. The benchmark’s own reported performance is worst precisely where the Pampas deployment needs it most: open-ended flood accuracy drops to 0.73 [Q90] (lowest disaster type), Water & Flood Characteristics MCQ F1 is ~0.63 [Q85] (lowest category), and Access & Inaccessibility binary is ~0.72–0.84 [Q82]. Geographic performance is reported at the coarse ‘South America’ regional level [Q97], with no Argentina-specific breakdown. There is no evaluation of Spanish-language output, no evaluation of output suitable for consumption by Rioplatense Spanish-speaking emergency managers, and no evaluation of output alignment with SINAGIR/AFE reporting formats. The benchmark authors themselves explicitly warn against treating performance as deployment-readiness [Q113, Q114, Q122], which is directly at odds with the operational deployment framing of the use case.

ID	Evidence
Q54	“To ensure reproducibility and fair comparison, all experiments are conducted with the temperature parameter set to zero, yielding deterministic outputs.” (p.4)
Q65	“Because model outputs are often inconsistently formatted, we applied an LLM-based post-processing step using GPT-4o to standardize responses across question types.” (p.5)
Q66	“This step extracts clean, structured outputs: binary questions are normalized to exactly Yes or No, while multiple-choice responses are converted into a JSON array containing the selected option(s).” (p.5)
Q74	“We measure three metrics in our evaluations, one for each question type: Binary accuracy, Open-ended accuracy, and Multiple-choice F1-score.” (p.5)
Q75	“Binary accuracy: The fraction of binary questions where the model’s Yes/No prediction exactly matches the ground truth over the total number of predictions.” (p.5)
Q76	“Open-ended accuracy: The percentage of open-ended questions where the model’s output is rated as Correct by the LLM-as-a-judge.” (p.5)
Q77	“Multiple-choice F1-score: We evaluate all MCQs jointly as a multi-label classification task.” (p.5)
Q82	“Binary accuracy is high across categories: Environmental/Terrain Context is perfect (≈ 1.00) and Built Environment Damage is consistently strong (≈ 0.93). The hardest binary category is Access & Inaccessibility (range ≈ 0.72–0.84), with Movement Restrictions & Controls (≈ 0.80) and Water & Flood Characteristics (≈ 0.83) also below the top tier.” (p.6)
Q85	“The weakest category is Water & Flood Characteristics (avg F1 ≈ 0.63), and Access & Inaccessibility is also challenging (avg F1 ≈ 0.72), where some models over-select options (higher false positives).” (p.6)
Q90	“the accuracy shows the largest spread, from a low of 0.73 on Floods to a peak of 0.94 on Landslides (GPT-4.1-mini). Accidents and Earthquakes sit in the low-to-mid 0.80s, whereas Fires drop to 0.79 and Hurricanes actually climb to 0.88.” (p.6)
Q97	“Open-source models are competitive in well-performing regions (e.g., South America and Southeast Asia), yet drop more sharply in the Middle East where all models compress toward the lower end of the scale.” (p.7)
Q113	“Performance metrics should not be treated as indicators of deployment readiness, as errors in disaster-related perception and reasoning can have significant societal consequences.” (p.8)
Q114	“DisasterVQA is therefore best understood as a controlled diagnostic tool for identifying strengths, failure modes, and research gaps in current vision–language models, rather than as a proxy for real-world disaster response performance.” (p.8)

Q122	"The benchmark is provided to advance responsible AI research and should be interpreted strictly as an evaluation resource rather than a measure of real-world readiness." (p.9)
Q123	"We did not report error bars because all models were evaluated under deterministic settings (temperature set to zero), resulting in identical outputs across repeated runs." (p.11)

Strengths

- Three complementary metric types (binary, MCQ F1, open-ended) [\[Q74\]](#) provide multi-faceted evaluation useful for comparing candidate VLMs
- Deterministic settings [\[Q54, Q123, Q124\]](#) ensure reproducible comparisons across models
- LLM-judge normalization allows semantic equivalence for open-ended answers [\[Q67, Q68\]](#), reducing false negatives from formatting variations
- Regional performance heatmaps [\[Q95\]](#) provide at least a coarse view of geographic variation
- Multi-label MCQ format [\[Q77\]](#) captures ambiguity present in real flood scenes

ID	Evidence
Q54	"To ensure reproducibility and fair comparison, all experiments are conducted with the temperature parameter set to zero, yielding deterministic outputs." (p.4)
Q67	"For open-ended questions, we adopt an LLM-as-a-judge approach with GPT-4o to assess answer correctness in addition to output normalization." (p.5)
Q68	"The judge model compares the extracted response against the ground-truth answer, accounting for semantic equivalence, paraphrasing, and formatting variations, and produces a structured decision (Correct or Incorrect) along with the normalized answer." (p.5)
Q74	"We measure three metrics in our evaluations, one for each question type: Binary accuracy, Open-ended accuracy, and Multiple-choice F1-score." (p.5)
Q77	"Multiple-choice F1-score: We evaluate all MCQs jointly as a multi-label classification task." (p.5)
Q95	"Figure 7 presents three heatmaps of model performance across different geographic regions. There are several pat-" (p.6)
Q123	"We did not report error bars because all models were evaluated under deterministic settings (temperature set to zero), resulting in identical outputs across repeated runs." (p.11)
Q124	"This design choice ensures reproducibility and enables fair, seed-independent comparisons between models." (p.11)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Expected output modality is text — matches the general form Pampas emergency managers would consume in a decision-support interface. However, output is in English only [dataset schema; all examples in registry]; deployment requires Rioplatense Spanish. This is a fundamental form mismatch.

OF-2: Not applicable — the deployment does not require speech output; emergency managers work at desktop/mobile interfaces [regional YAML: device_context]. However, the benchmark also does not evaluate text-to-speech, so this dimension is neutral.

OF-3: Literacy is very high in Argentina, and emergency managers have tertiary/university education [regional YAML]; the literacy dimension is not a limiting factor. The relevant accessibility concern is Spanish-language output, not literacy.

OF-4: External validity harms: (a) English-only output means benchmark performance does not predict Spanish-output quality; (b) flood-specific performance is worst in the categories most relevant to the deployment [Q85, Q90]; (c) no Argentina-specific geographic breakdown [Q97]; (d) benchmark authors explicitly disclaim deployment-readiness interpretation [Q113, Q122] but the use case treats it as operational; (e) minor: single split with no held-out test set limits external-validity claims [dataset schema].

ID	Evidence
Q74	"We measure three metrics in our evaluations, one for each question type: Binary accuracy, Open-ended accuracy, and Multiple-choice F1-score." (p.5)
Q85	"The weakest category is Water & Flood Characteristics (avg F1 $\approx$ 0.63), and Access & Inaccessibility is also challenging (avg F1 $\approx$ 0.72), where some models over-select options (higher false positives)." (p.6)
Q90	"the accuracy shows the largest spread, from a low of 0.73 on Floods to a peak of 0.94 on Landslides (GPT-4.1-mini). Accidents and Earthquakes sit in the low-to-mid 0.80s, whereas Fires drop to 0.79 and Hurricanes actually climb to 0.88." (p.6)
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Information Gaps

- No Argentina-specific geographic performance breakdown [Q97 only reports 'South America' aggregate]
- No documented evaluation of any model's Spanish-language output

Requires Expert Verification

- Whether the observed flood-specific performance gaps (0.73 open-ended, 0.63 MCQ F1) meet minimum operational thresholds for Argentine EOC decision support

Remediation

Gap 1: Documented worst-in-benchmark performance on flood-specific tasks (open-ended 0.73, MCQ F1 0.63); no Argentina-specific geographic breakdown; benchmark authors disclaim deployment-readiness interpretation

Recommendation: Do not treat aggregate DisasterVQA scores as operational readiness indicators. Report flood-specific and Argentina-specific (once curated) subset metrics separately, and establish minimum accuracy thresholds for each SINAGIR-mapped output category before deployment. Include a rejection/uncertainty output form so models can defer to human judgment on low-confidence cases.